```
========== GENERAL TREE MENU ==========
1. Create root
2. Add child to a node
3. Display tree
4. Display leaf and non-leaf nodes (with count)
5. Count nodes at each level
6. Query children of a node
7. Load sample tree
8. Exit
Enter choice: 7
Sample tree loaded.

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8. Exit
Enter choice: 3

----- Tree Structure -----
1
    |-- 2
        |-- 5  (leaf)
        |-- 6
            |-- 11  (leaf)
            |-- 12  (leaf)
    |-- 3
        |-- 7  (leaf)
    |-- 4
        |-- 8  (leaf)
        |-- 9  (leaf)
        |-- 10  (leaf)

========== GENERAL TREE MENU ==========
1. Create root
```

```
        |-- 10  (leaf)

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Enter choice: 4

Leaf nodes (7): 5 11 12 7 8 9 10
Non-leaf nodes (5): 1 2 6 3 4
Total nodes: 12

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7. Load sample tree
8. Exit
Enter choice: 5

----- Nodes at Each Level -----
Level 0 : 1 node(s) -> 1
Level 1 : 3 node(s) -> 2 3 4
Level 2 : 6 node(s) -> 5 6 7 8 9 10
Level 3 : 2 node(s) -> 11 12
Height of tree (levels): 4

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```